

Code No: 157GD**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech IV Year I Semester Examinations, December-2023/January-2024****DATABASE SECURITY****(Computer Science and Engineering - Data Science)****Time: 3 Hours****Max.Marks:75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) List security threats for databases. [2]
- b) What is meant by system availability? [3]
- c) What is the main drawback of low-watermark policy for subjects? [2]
- d) What access modes are executable by subjects? [3]
- e) Give example for exact compromise. [2]
- f) What is a well-formed transaction? [3]
- g) What is implicit authorization? [2]
- h) Give an example of a role lattice. [3]
- i) How to store rules in active database? [2]
- j) What is the role of triggers in active database? [3]

PART – B**(50 Marks)**

2. Illustrate the main inference channels in a database system and inference control. [10]
- OR**
3. Compare and contrast access matrix model with PN model. [10]
4. Explain how the Jajodia and Sandhu model formalizes multilevel relations and the operations executable on them. [10]
- OR**
5. Discuss in detail the basic security features of UNIX operating system. [10]
6. What are the main security requirements a database management system should provide? Explain them with illustrations. [10]
- OR**
7. Describe the architecture of Real Time Security System (RETISS) for threat detection.[10]
8. Review the main proposals to enforce security in object oriented DBMS. [10]
- OR**
9. Explain SORION model as an extension to the ORION object-oriented model. [10]
10. Describe Jajodia and Kogan's model for protecting active databases. [10]
- OR**
11. Demonstrate the working of MLS rule system in active databases. [10]